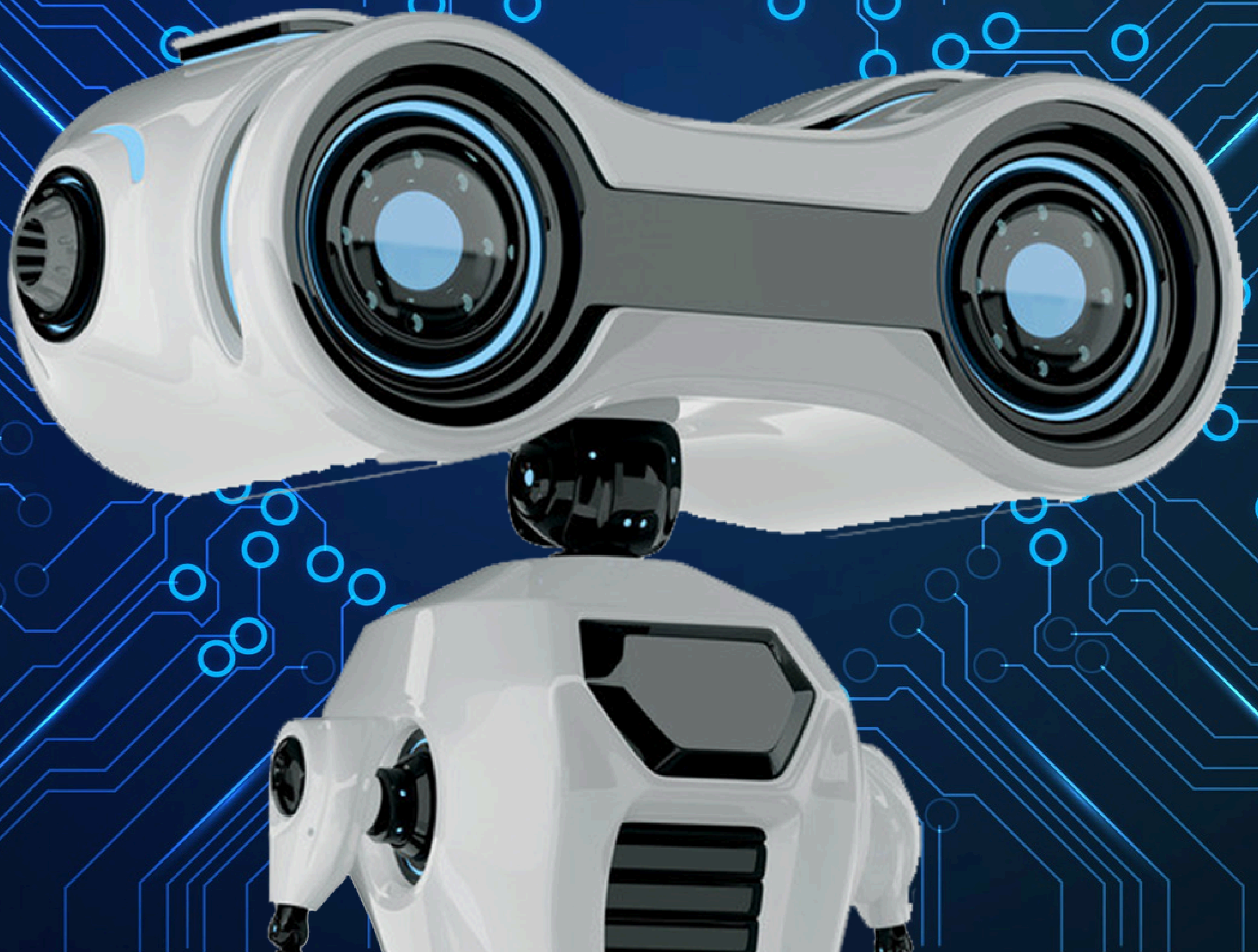
Video Game AI - Decision Tree CRUD for NPC Behaviour



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Introduction :

- NPCs (Non-Player Characters) make decisions in games

Actions include:

- Attack
- Patrol
- Run
- Search
- Decision Trees help simulate intelligent behaviour

Problem statement:

NPCs need smart decision-making

Manual logic is complex

Need system to:

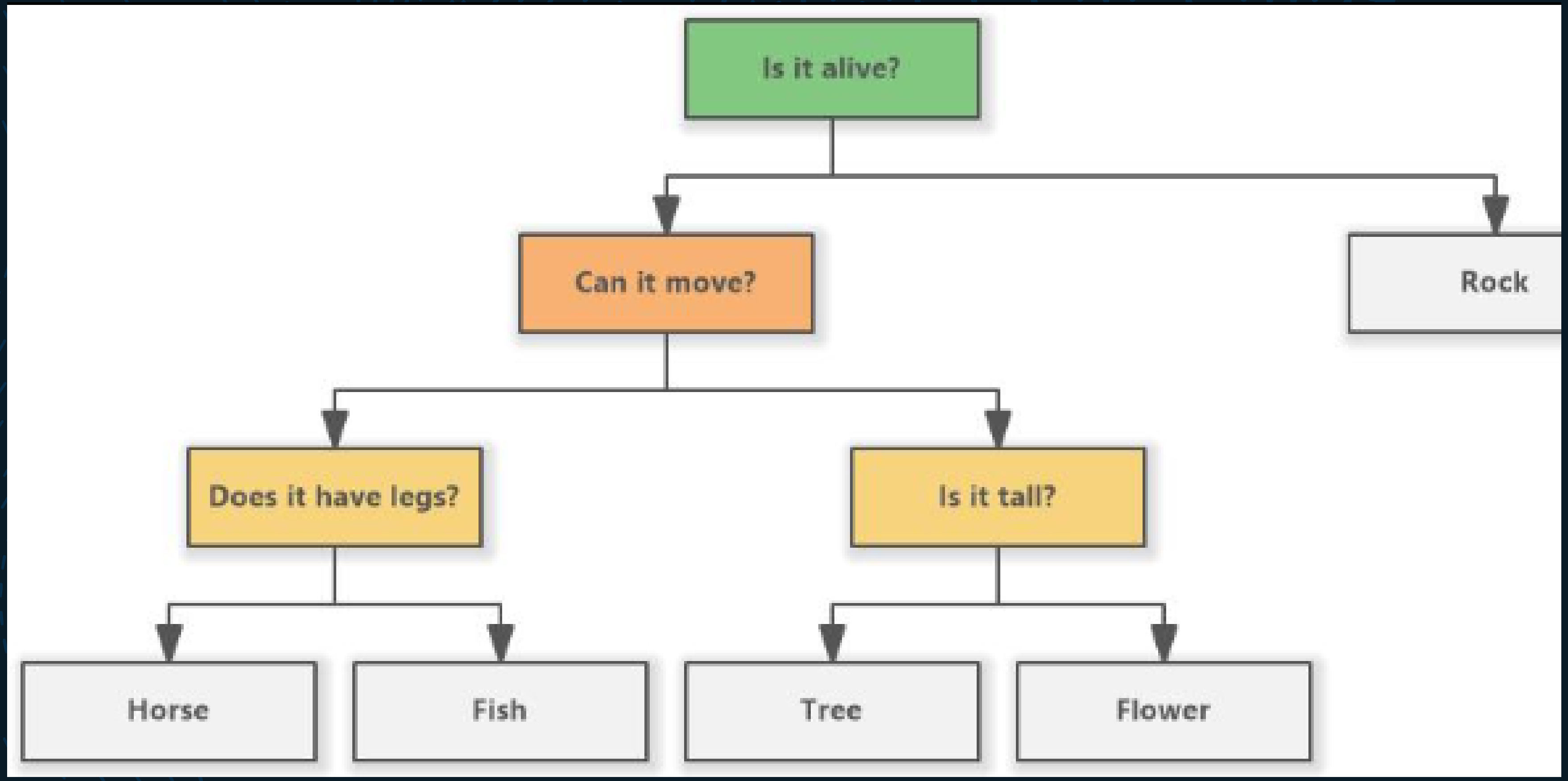
Add decisions

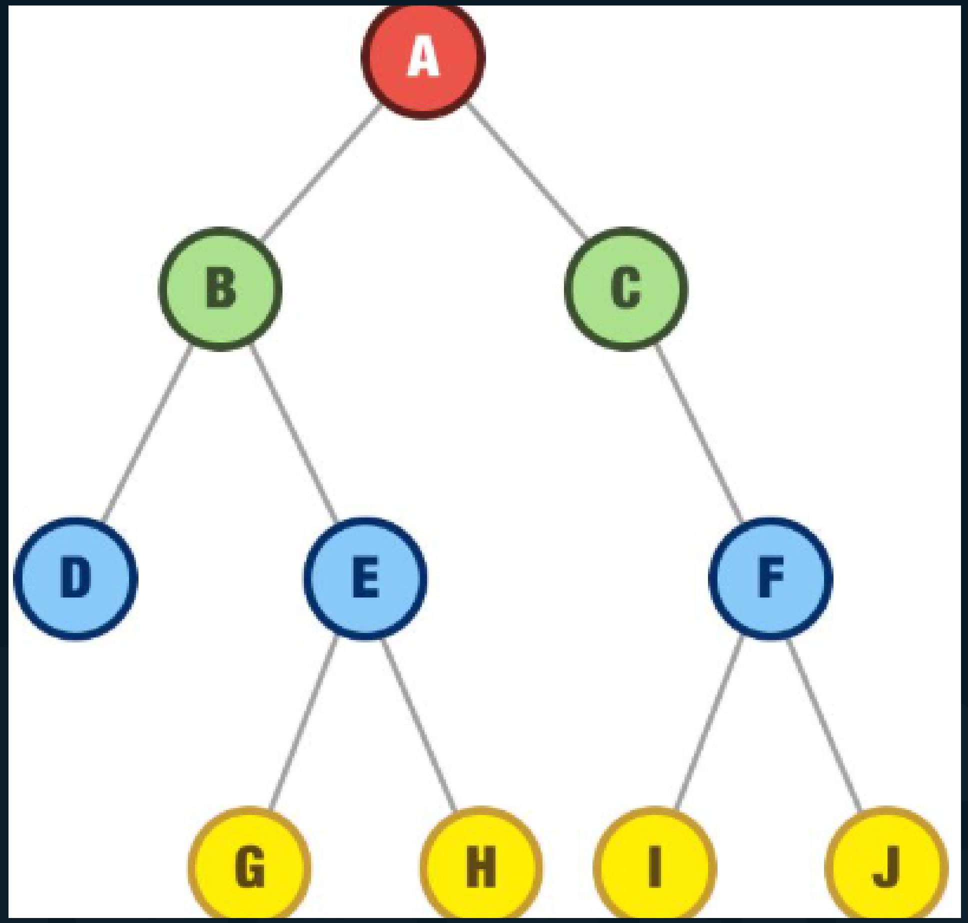
Update decisions

Delete decisions

Search decisions

Solution: Decision Tree + CRUD Operation





Data Structure Used:

Binary Tree

Node contains:

ID

Decision

Left child

Right child

Node Structure(Code)

```
struct Node
{ int id;
char decision[100];
struct Node *left;
struct Node *right;
      };
```

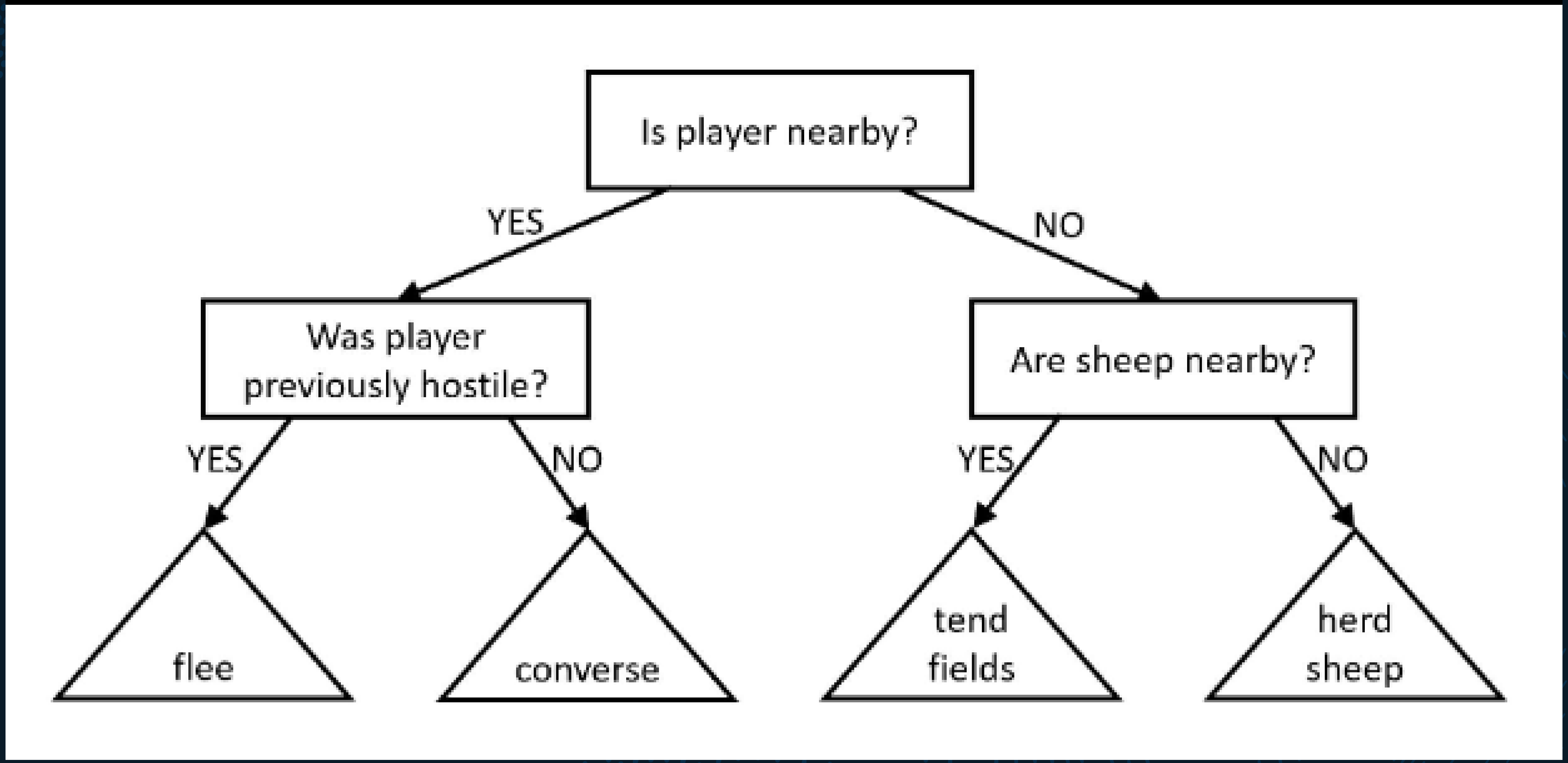
System Design

Example Decision Tree:

Enemy Nearby?

Yes → Attack Enemy

No → Patrol Area



CRUD Operations

Create → Add Node

Read → Display

Tree Update → Modify

Decision Delete → Remove Node

Algorithm (Insert)

1. Start
2. Check if tree empty
3. Create node
4. Compare IDs
5. Insert left or right

Algorithm (Search)

1.Compare ID with root

2.If equal → Found

3.If smaller → Left

4.If larger → Right Algorithm (Update & Delete)

Update:

Search node

Modify decision

Delete:

Find node

Remove it

Adjust tree

Implementation Details :

C Programming Structures

malloc & free

Functions

Tree traversal

Menu Operations

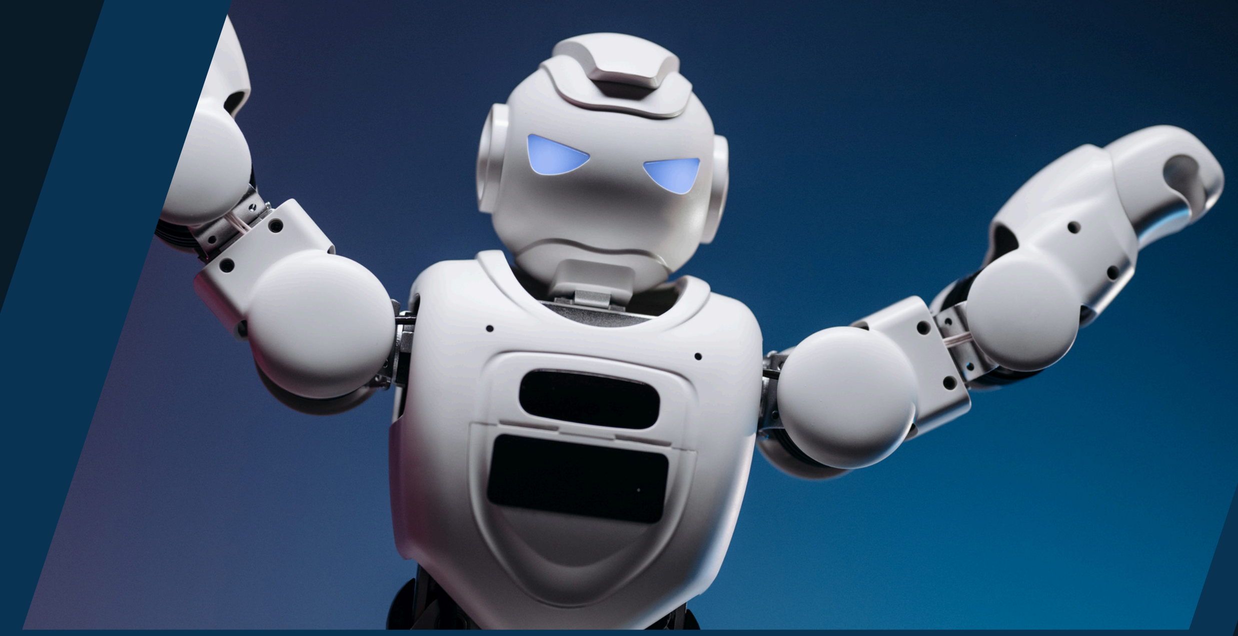
Add Node

Delete Node

Update Node

Search Node

Display Tree Exit



Limitations :

Console-based only

No graphical view

Basic AI logic

Future Enhancements

Graphical tree visualization

Game engine integration

Advanced AI logic

Multiple conditions

Conclusion :

Implemented Decision Tree

CRUD operations successful

Useful for Game AI

Learned data structures & memory

Thank you